

ECE 332 — Homework 4: Generated Answers from HW4 Cheat Sheet

Each problem: state the goal equation $\rightarrow$ identify missing variables $\rightarrow$ solve each $\rightarrow$ plug back $\rightarrow$ box the answer.

Symbol reference (used throughout):

- l — antenna physical length (m)
- $\lambda = c/f$ — wavelength (m); $c = 3 \times 10^8$ m/s
- I_0 — peak current at antenna feed (A)
- R — distance from antenna to observation point (m)
- θ — angle from antenna axis (rad or deg)
- $\eta_0 = 120\pi \approx 377 \Omega$ — intrinsic impedance of free space
- $k = 2\pi/\lambda$ — wavenumber (rad/m)
- $S(R, \theta)$ — average power density radiated (W/m²)
- $F(\theta) = S(\theta)/S_{\max}$ — normalized radiation intensity (unitless)
- β — half-power beamwidth (rad or deg)
- Ω_p — pattern solid angle (sr)
- $D = 4\pi/\Omega_p$ — directivity (unitless, often dB)
- $A_e = \lambda^2 D / (4\pi)$ — effective aperture area (m²)
- A_p — physical cross-section area (m²)
- G_t, G_r — antenna gains (linear)
- P_t, P_r — transmitted / received power (W)
- $k_B = 1.38 \times 10^{-23}$ J/K — Boltzmann constant
- T_{sys} — system noise temperature (K)
- B — receiver bandwidth (Hz)

Key shortcuts: Hertzian dipole ($l/\lambda < 1/50$) $\Rightarrow D = 1.5$. Half-wave dipole ($l = \lambda/2$) $\Rightarrow D = 1.64$, $R_{\text{rad}} = 73.2 \Omega$. 3 dB $\Rightarrow$ linear gain of 2.

Problem 1 — Hertzian Dipole Power Density at 5 km

8 pts

Setup: $l = 1$ m, $f = 1$ MHz, $I_0 = 12$ A, $R = 5$ km, $\theta = 45^\circ$. Find $S(R, \theta)$.

DIPOLE TYPE BY LENGTH
 $l = \text{antenna rod length (m)}$

$$\lambda \text{ (wavelength)} = \frac{c}{f} = \frac{3 \times 10^8}{f} \text{ (m)}$$

l/λ	Type	Use formula
$< 1/50$	Hertzian (short)	HERTZIAN $S(R, \theta)$, $D = 1.5$
$= 1/4$	Quarter-wave	ARB formula, $\pi l/\lambda = \pi/4$
$= 1/2$	Half-wave	half-wave shortcut, $D = 1.64$, $R_{\text{rad}} = 73 \Omega$
$= 1$	Full-wave	ARB, $D = 2.41$, $R_{\text{rad}} = 199 \Omega$
other	Arbitrary	ARB formula

COMMON DIPOLE PARAMETERS

Lossless ($\xi = 1$): $G = D$.

Type	l	D	D_{dB}	$R_{\text{rad}} (\Omega)$
Isotropic	N/A	1	0	—
Hertzian	$< \lambda/50$	1.5	1.76	$80\pi^2(l/\lambda)^2$
Half-wave	$\lambda/2$	1.64	2.15	73.2
Monopole	$\lambda/4$ (gnd)	1.64	2.15	36.6
Full-wave	λ	2.41	3.82	199

Half-wave: $P_{\text{rad}} = 36.6 I_0^2 \text{ W}$. Monopole ($\lambda/4$ above ground): same patt. as half-wave but $P_{\text{rad}}, R_{\text{rad}}$ halved.

HERTZIAN DIPOLE — $S(R, \theta)$

 Far-field E,H phasors *small $l \ll \lambda$*

$$\tilde{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi R} e^{-jkR} \sin \theta, \quad \tilde{H}_\phi = \tilde{E}_\theta / \eta_0$$

 Power density formulas $l/\lambda < 1/50$; *far field*

$$\text{Power density} \quad S(R, \theta) = S_{\text{max}} \sin^2 \theta \text{ (W/m}^2\text{)}$$

$$\text{Max pwr density} \quad S_{\text{max}} = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2} \text{ (W/m}^2\text{)}$$

l : rod length (m)	I_0 : peak current (A)
R : obs. distance (m)	θ : angle from dipole axis (rad)
$k = 2\pi/\lambda$ (rad/m)	$\eta_0 = 120\pi \approx 377 \Omega$

$$\text{Total radiated power} \quad P_{\text{rad}} = \frac{\eta_0 \pi}{3} (I_0 l / \lambda)^2 \text{ (W)}$$

Step 0 — Classify the dipole. From DIPOLE TYPE BY LENGTH, we need l/λ .

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{10^6} = 300 \text{ m}, \quad \frac{l}{\lambda} = \frac{1}{300} < \frac{1}{50} \Rightarrow \textbf{Hertzian}$$

Goal: find $S(R, \theta)$ using the Hertzian power density formula (from HERTZIAN DIPOLE box):

$$S(R, \theta) = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2} \sin^2 \theta$$

Need: $\eta_0, k, I_0, l, R, \theta$.

Solve each unknown:

η_0 : constant, free-space impedance.

$$\eta_0 = 120\pi \Omega$$

k : wavenumber from wavelength.

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{300} \text{ rad/m}$$

I_0, l, R, θ : given.

$$I_0 = 12 \text{ A}, \quad l = 1 \text{ m}, \quad R = 5000 \text{ m}, \quad \theta = 45^\circ$$

Plug back in:

$$S = \frac{(120\pi)(2\pi/300)^2(12)^2(1)^2}{32\pi^2(5000)^2} \sin^2(45^\circ)$$

From COMMON ANGLES box: $\sin^2(45^\circ) = 1/2$.

Simplify $k^2 = (2\pi/300)^2$:

$$(2\pi/300)^2 = \frac{4\pi^2}{90000} = \frac{4\pi^2}{9 \times 10^4}$$

Simplify $I_0^2 = 144, l^2 = 1, R^2 = (5000)^2 = 25 \times 10^6$.

Substitute everything:

$$S = \frac{(120\pi) \cdot (4\pi^2/9 \times 10^4) \cdot 144 \cdot 1}{32\pi^2 \cdot 25 \times 10^6} \cdot \frac{1}{2}$$

Group the numerical coefficients (separate from π powers):

Numerator: $120 \cdot 4 \cdot 144 / (9 \times 10^4) = 69120 / 9 \times 10^4 = 0.7680$

With π powers: $\pi \cdot \pi^2 = \pi^3$, so numerator = $0.7680 \pi^3$

Denominator: $32 \cdot 25 \times 10^6 = 800 \times 10^6 = 8 \times 10^8$

With π powers: π^2 , so denominator = $8 \times 10^8 \pi^2$

Combine:

$$\begin{aligned} S &= \frac{0.7680 \pi^3}{8 \times 10^8 \pi^2} \cdot \frac{1}{2} = \frac{0.7680 \pi}{8 \times 10^8} \cdot \frac{1}{2} \\ &= \frac{0.7680 \cdot 3.14159}{8 \times 10^8} \cdot 0.5 = \frac{2.413}{8 \times 10^8} \cdot 0.5 = \frac{1.207}{8 \times 10^8} \\ &= 1.508 \times 10^{-9} \text{ W/m}^2 \end{aligned}$$

$S(5 \text{ km}, 45^\circ) = 1.51 \times 10^{-9} \text{ W/m}^2 = 1.51 \text{ nW/m}^2$

Sanity: 12 A at 1 MHz through 1 m radiates 1.5 nW/m² at 5 km. The $1/R^2$ falloff dominates — most power scatters into space, only a tiny fraction reaches any one point.

Problem 2 — Beam Parameters from $F(\theta) = e^{-20\theta^2}$ **15 pts****Setup:** $F(\theta) = \exp(-20\theta^2)$ for $0 \leq \theta \leq \pi$ (θ in radians). Find (a) HPBW β , (b) Ω_p , (c) D .**BEAM PARAMETERS — β , Ω_p , D** **Normalize** *always start here*

$$F(\theta) = S(\theta)/S_{\max} \text{ (unitless)}$$

Beamwidth β at threshold X Set $F(\theta_X) = X$, solve for θ_X ; $\beta = 2\theta_X$ (symmetric)

Threshold	$X = 10^{\text{dB}/10}$	θ_X
HPBW (−3 dB)	0.5	$\sqrt{\frac{\ln 2}{a}}$
−6 dB	0.25	$\sqrt{\frac{\ln 4}{a}}$
−10 dB	0.1	$\sqrt{\frac{\ln 10}{a}}$
any $F(\theta_X)$	any X	$F(\theta_X) = X$

Pattern solid angle Ω_p *recognize pattern, look up*

$$\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin \theta \, d\theta \, d\phi \text{ (sr)}$$

$$\text{No } \phi \text{ dep.: } \Omega_p = 2\pi \int_0^\pi F(\theta) \sin \theta \, d\theta.$$

 θ is the integration variable — DON'T plug in a number for θ (e.g. $\theta_{1/2}$). Integrate over $\theta \in [0, \pi]$.

Pattern $F(\theta)$	Ω_p (sr)
Isotropic ($F=1$)	$4\pi \approx 12.57$
Hertzian $\sin^2 \theta$	$8\pi/3 \approx 8.38$
Half-wave dipole	≈ 7.66
Narrow Gauss $e^{-a\theta^2}$	$\pi/a = \pi\theta_{1/2}^2 / \ln 2$
2-axis HPBW (aperture)	$\beta_{xz} \cdot \beta_{yz}$
Other (use integral)	numerical

*TI-36X: $[2nd][d/dx]$ for $\int_{\square}^{\square}$, RAD mode, multiply by 2π .***Directivity D** *always $D = 4\pi/\Omega_p$; common shortcuts below*

$$D = \frac{4\pi}{\Omega_p} \text{ (unitless)} \quad D_{\text{dB}} = 10 \log_{10} D$$

*WORD TRAP: “direction” = $\theta_{\max}$ (rad); “directivity” = D (unitless).***Gain (if asked)** ξ = radiation efficiency (unitless), $0 < \xi \leq 1$

$$G = \xi D \text{ (unitless)} \quad G_{\text{dB}} = 10 \log_{10} G \text{ (dB)}$$

Lossless / ideal antenna $\Rightarrow G = D$.

$$\xi = \frac{R_{\text{rad}}}{R_{\text{rad}} + R_{\text{loss}}} = \frac{P_{\text{rad}}}{P_{\text{rad}} + P_{\text{loss}}}; \quad R_{\text{rad}} = 2P_{\text{rad}}/I_0^2 \text{ (}\Omega\text{)};$$

$$R_{\text{in}} = R_{\text{rad}} + R_{\text{loss}}.$$

(a) Half-power beamwidth β (5 pts)**Goal:** find β using the HPBW procedure (BEAM PARAMETERS Step 2):

$$\text{Set } F(\theta) = 0.5, \text{ solve for } \theta \rightarrow \theta_{1/2}$$

$$\beta = 2\theta_{1/2} \text{ (symmetric beam)}$$

Need: $\theta_{1/2}$, found by setting our specific $F(\theta) = 0.5$.**Solve $\theta_{1/2}$:** plug $F = \exp(-20\theta^2)$:

$$\exp(-20\theta_{1/2}^2) = 0.5 \Rightarrow -20\theta_{1/2}^2 = \ln(0.5) = -0.693$$

$$\theta_{1/2}^2 = \frac{0.693}{20} = 0.03466, \quad \theta_{1/2} = \sqrt{0.03466} = 0.186 \text{ rad}$$

That's the half-angle from peak to one -3 dB point. Since the beam is symmetric about $\theta = 0$, the *full* half-power beamwidth is twice that:

$$\beta = 2\theta_{1/2} = 2(0.186) = 0.372 \text{ rad}$$

Convert rad $\rightarrow$ deg (multiply by $180/\pi$, from COMMON ANGLES box):

$$\beta = 0.372 \cdot \frac{180}{\pi} = 0.372 \cdot 57.296 = 21.31^\circ$$

$$\boxed{\beta = 0.372 \text{ rad} = 21.31^\circ}$$

(b) Pattern solid angle Ω_p (5 pts)

Goal: find Ω_p using the integral definition (BEAM PARAMETERS Step 3):

$$\Omega_p = \iint_{4\pi} F(\theta) \sin \theta \, d\theta \, d\phi$$

Need: evaluate this integral with our specific $F(\theta) = e^{-20\theta^2}$.

Set up the integral: F has no ϕ dependence, so $\int_0^{2\pi} d\phi = 2\pi$:

$$\Omega_p = 2\pi \int_0^\pi e^{-20\theta^2} \sin \theta \, d\theta$$

Solve numerically on TI-36X Pro (no closed form):

1. [mode] $\rightarrow$ **RAD**
2. [2nd] [d/dx] for $\int_{\square}^{\square}$ template
3. Lower limit: 0; Upper limit: $[\pi]$
4. Integrand: $e^{(-20 \times x^2)} \times \sin(x)$
5. Variable: x ; [enter] $\rightarrow$ returns ≈ 0.024792
6. Multiply: $\times [\pi] \times 2$ [enter] $\rightarrow \approx 0.15578$

$$\boxed{\Omega_p \approx 0.1558 \text{ sr}}$$

Keep the unrounded value for part (c). Textbook rounds to 0.156 first and gets $D = 80.55$; using full precision gives $D \approx 80.67$.

(c) Directivity D (5 pts)

Goal: find D using the standard formula (BEAM PARAMETERS Step 4):

$$D = \frac{4\pi}{\Omega_p}$$

Need: Ω_p (from part b).

Plug in: $\Omega_p = 0.15578$ sr:

$$D = \frac{4\pi}{0.15578} = \frac{12.566}{0.15578}$$

$$\boxed{D = 80.67}$$

In dB: $D_{dB} = 10 \log_{10}(80.67) = 19.07$ dB.

Sanity: $D \approx 80$ is highly directive (much narrower than half-wave's $D = 1.64$). The narrow Gaussian-like beam ($\beta \approx 21^\circ$) concentrates radiation into a small solid angle, so directivity is large.

Problem 3 — $\lambda/4$ Dipole Radiation Pattern

15 pts

Setup: Dipole of length $l = \lambda/4$. Find (a) $\theta_{\max}$, (b) $S_{\max}$, (c) plot $F(\theta)$.

ARBITRARY-LENGTH DIPOLE — $S(\theta)$ Power density at R any l

$$S(\theta) = \frac{15I_0^2}{\pi R^2} \left[\frac{\cos\left(\frac{\pi l}{\lambda} \cos \theta\right) - \cos\left(\frac{\pi l}{\lambda}\right)}{\sin \theta} \right]^2$$

At $\theta = 0, \pi$: bracket is 0/0. $L'Hôpital \Rightarrow S = 0$ (no axial radiation). TI-36X alt: evaluate bracket at $\theta = 0.001$ rad $\rightarrow$ tiny $\rightarrow 0$.

Normalized pattern dimensionless $F(\theta) = S(\theta)/S_{\max}$

COMMON DIPOLE PARAMETERS

Lossless ($\xi=1$): $G=D$.

Type	l	D	D_{dB}	$R_{\text{rad}} (\Omega)$
Isotropic	N/A	1	0	—
Hertzian	$< \lambda/50$	1.5	1.76	$80\pi^2(l/\lambda)^2$
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Half-wave: $P_{\text{rad}} = 36.6 I_0^2$ W. Monopole ($\lambda/4$ above ground): same patt. as half-wave but $P_{\text{rad}}, R_{\text{rad}}$ halved.

COMMON ANGLES (deg $\leftrightarrow$ rad, trig values)Conversion multiply by $\pi/180$ or $180/\pi$

$$\theta_{\text{rad}} = \theta_{\text{deg}} \cdot \frac{\pi}{180} \quad \theta_{\text{deg}} = \theta_{\text{rad}} \cdot \frac{180}{\pi}$$

°	rad	$\sin \theta$	$\cos \theta$	$\sin^2 \theta$	$\cos^2 \theta$
0	0	0	1	0	1
30	$\pi/6$	1/2	$\sqrt{3}/2$	1/4	3/4
45	$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	1/2	1/2
60	$\pi/3$	$\sqrt{3}/2$	1/2	3/4	1/4
90	$\pi/2$	1	0	1	0
180	π	0	-1	0	1

Antenna formulas usually take θ in rad. RAD mode on calc.

Common setup for all 3 parts.

Goal: starting point for parts (a)-(c) is the arbitrary-length dipole power density (from ARBITRARY-LENGTH DIPOLE box):

$$S(\theta) = \frac{15I_0^2}{\pi R^2} \left[\frac{\cos\left(\frac{\pi l}{\lambda} \cos \theta\right) - \cos\left(\frac{\pi l}{\lambda}\right)}{\sin \theta} \right]^2$$

Need: plug in $l = \lambda/4$ to simplify.

Substitute $l = \lambda/4 \Rightarrow \pi l/\lambda = \pi/4$:

$$S(\theta) = \frac{15I_0^2}{\pi R^2} \left[\frac{\cos\left(\frac{\pi}{4} \cos \theta\right) - \cos(\pi/4)}{\sin \theta} \right]^2$$

From COMMON ANGLES table: $\cos(\pi/4) = \sqrt{2}/2 = 0.7071$.

$$S(\theta) = S_0 \left[\frac{\cos\left(\frac{\pi}{4} \cos \theta\right) - \frac{\sqrt{2}}{2}}{\sin \theta} \right]^2, \quad S_0 \equiv \frac{15I_0^2}{\pi R^2}$$

(a) Direction of maximum radiation $\theta_{\max}$ (5 pts)

Goal: find $\theta_{\max}$ where $S(\theta)$ is maximum. **Need:** know the shape of $S(\theta)$ over $\theta \in [0, \pi]$.
 Three-step argument: endpoints, symmetry, sampling.

Step (i) — endpoints via L'Hôpital. Define $f(\theta) = \cos(\frac{\pi}{4} \cos \theta) - \frac{\sqrt{2}}{2}$ and $g(\theta) = \sin \theta$. At $\theta = 0$:

$$f(0) = \cos(\pi/4 \cdot 1) - \sqrt{2}/2 = \sqrt{2}/2 - \sqrt{2}/2 = 0$$

$$g(0) = \sin(0) = 0 \Rightarrow 0/0 \text{ form, apply L'Hôpital.}$$

Derivatives (chain rule for f):

$$f'(\theta) = \frac{\pi}{4} \sin(\frac{\pi}{4} \cos \theta) \sin \theta, \quad g'(\theta) = \cos \theta$$

$$\lim_{\theta \rightarrow 0} \frac{f'(\theta)}{g'(\theta)} = \frac{(\pi/4) \sin(\pi/4) \cdot 0}{1} = 0$$

So $S(0)/S_0 = 0^2 = 0$. Same algebra at $\theta = \pi$: $S(\pi) = 0$. **Peak is in $(0, \pi)$.**

Step (ii) — symmetry. Substitute $\theta \rightarrow \pi - \theta$: $\cos(\pi - \theta) = -\cos \theta$, but $\cos$ is even so $\cos((\pi/4) \cos(\pi - \theta)) = \cos((\pi/4) \cos \theta)$. And $\sin(\pi - \theta) = \sin \theta$. So $S(\theta) = S(\pi - \theta)$. **Pattern mirrors about $\theta = \pi/2$.**

Step (iii) — sample to confirm single peak.

At $\theta = 30^\circ$: $\cos(30^\circ) = 0.866$, $(\pi/4)(0.866) = 0.680$, $\cos(0.680) = 0.777$, $0.777 - 0.707 = 0.070$, $\sin(30^\circ) = 0.500$, $0.070/0.500 = 0.140$, $(0.140)^2 = \mathbf{0.020}$.

At $\theta = 60^\circ$: $\cos(60^\circ) = 0.500$, $(\pi/4)(0.500) = 0.393$, $\cos(0.393) = 0.924$, $0.924 - 0.707 = 0.217$, $\sin(60^\circ) = 0.866$, $0.217/0.866 = 0.250$, $(0.250)^2 = \mathbf{0.063}$.

At $\theta = 90^\circ$: $\cos(90^\circ) = 0$, $(\pi/4)(0) = 0$, $\cos(0) = 1$, $1 - 0.707 = 0.293$, $\sin(90^\circ) = 1$, $0.293/1 = 0.293$, $(0.293)^2 = \mathbf{0.0858}$.

θ	30°	60°	90°
S/S_0	0.020	0.063	0.0858

Monotone rise to 90° , then symmetric fall (by step (ii)). Single peak at the center.

$$\theta_{\max} = \pi/2 = 90^\circ \text{ (broadside)}$$

(b) Expression for $S_{\max}$ (5 pts)

Goal: find $S_{\max}$ by evaluating S at the maximum direction:

$$S_{\max} = S(\theta_{\max})$$

Need: $\theta_{\max}$ (from part a: $\theta_{\max} = \pi/2$).

Plug $\theta = \pi/2$ into $S(\theta)$ step by step:

Inner cos: $\cos(\pi/2) = 0$, so $\frac{\pi}{4} \cos(\pi/2) = 0$.

Outer cos: $\cos(0) = 1$.

Denominator: $\sin(\pi/2) = 1$.

Bracket: $(1 - \sqrt{2}/2)/1 = 1 - \sqrt{2}/2 = 1 - 0.7071 = 0.2929$.

Square: $(0.2929)^2 = 0.0858$.

Multiply by S_0 :

$$S_{\max} = 0.0858 \cdot \frac{15I_0^2}{\pi R^2} \approx \frac{1.287 I_0^2}{\pi R^2}$$

(c) Normalized pattern $F(\theta) = S(\theta)/S_{\max}$ (5 pts)

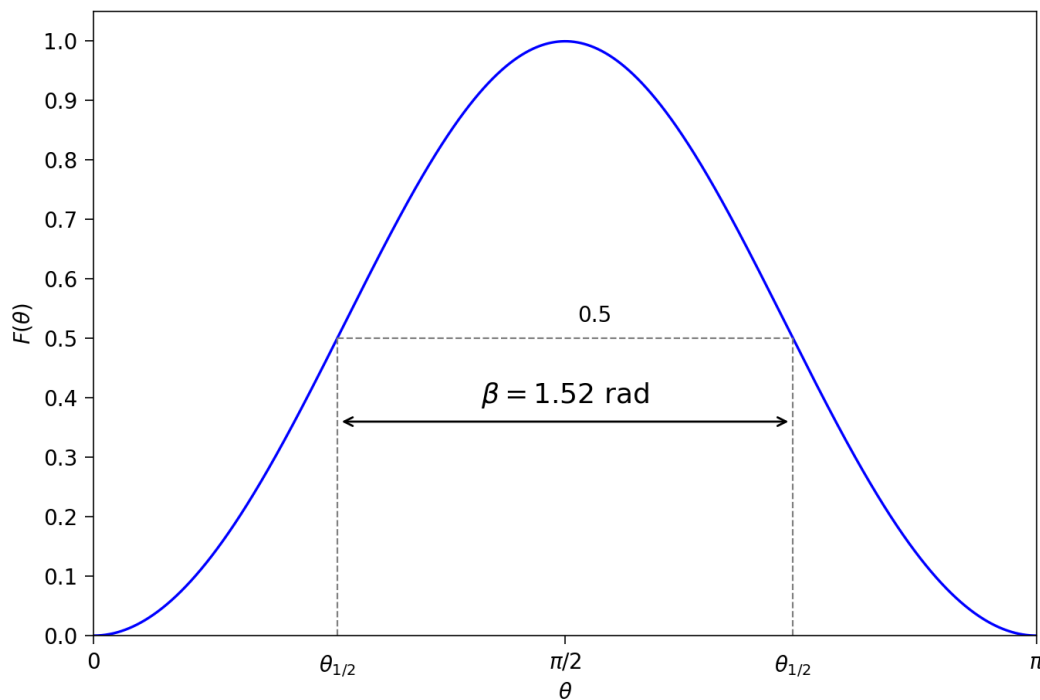
Goal: find $F(\theta)$ by dividing $S(\theta)$ by $S_{\max}$, then plot. **Need:** $S(\theta)$ and $S_{\max}$ (both from above).

Compute the normalization coefficient: $1/0.0858 = 11.66$.

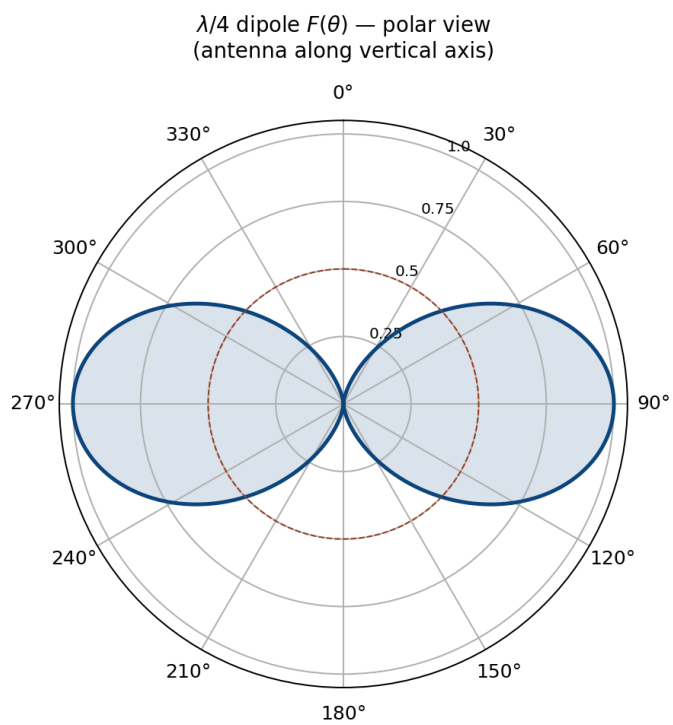
Form $F(\theta)$:

$$F(\theta) = 11.66 \left[\frac{\cos\left(\frac{\pi}{4} \cos \theta\right) - \frac{\sqrt{2}}{2}}{\sin \theta} \right]^2$$

Plot: computed in Python (`make_p3_plot.py`), shown below in both rectangular and polar form.



Bell-shaped curve peaking at $\theta = \pi/2$ with $F = 1$, dropping to 0 at $\theta = 0$ and π . Symmetric about $\theta = \pi/2$. Similar shape to half-wave dipole pattern but slightly narrower since the dipole is shorter.



Polar view shows the broadside donut pattern — strong perpendicular to the dipole, null along the axis. HPBW from the plot $\approx 87^\circ$.

Problem 4 — Effective Area vs. Physical Cross-section

5 pts

Setup: Half-wave dipole at $f = 100$ MHz, wire diameter $d = 2$ cm. Find A_e and compare to A_p .**EFFECTIVE AREA A_e** **Definition** *antenna's RX cross section (m^2)*

$$A_e = \frac{\lambda^2 D}{4\pi} \text{ (m}^2\text{)}$$

Physical cross section *wire dipole, diameter d*

$$A_p = l d = \frac{\lambda}{2} d \text{ (half-wave)}$$

Inverse: D from A_e

$$D = \frac{4\pi A_e}{\lambda^2} \text{ (unitless)}$$

Hertzian special case $D = 1.5$

$$A_e^{\text{Hertz}} = \frac{3\lambda^2}{8\pi} \approx 0.119\lambda^2 \text{ (m}^2\text{)}$$

*Thin wire: $A_e \gg A_p$. Antenna captures field over $\sim \lambda$.***COMMON DIPOLE PARAMETERS***Lossless ($\xi = 1$): $G = D$.*

Type	l	D	D_{dB}	$R_{\text{rad}} \text{ (}\Omega\text{)}$
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*Half-wave: $P_{\text{rad}} = 36.6 I_0^2$ W. Monopole ($\lambda/4$ above ground): same patt. as half-wave but $P_{\text{rad}}, R_{\text{rad}}$ halved.***Part 1 — Effective area A_e** **Goal:** find A_e using the standard formula (from EFFECTIVE AREA box):

$$A_e = \frac{\lambda^2 D}{4\pi}$$

Need: λ and D .**Solve each unknown:** λ : use $\lambda = c/f$.

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{10^8} = 3 \text{ m}$$

 D : look up half-wave in COMMON DIPOLE PARAMETERS table.

$$D_{\text{half-wave}} = 1.64$$

Plug back in:

$$A_e = \frac{(3)^2(1.64)}{4\pi} = \frac{9 \cdot 1.64}{4\pi} = \frac{14.76}{12.566}$$

$$A_e = 1.17 \text{ m}^2$$

Part 2 — Physical cross-section A_p **Goal:** find A_p using the wire-dipole formula:

$$A_p = l \cdot d$$

Need: l and d .**Solve each unknown:** l : half-wave means $l = \lambda/2$ (from COMMON DIPOLE PARAMS table).

$$l = \frac{\lambda}{2} = \frac{3}{2} = 1.5 \text{ m}$$

 d : given.

$$d = 2 \text{ cm} = 0.02 \text{ m}$$

Plug back in:

$$A_p = (1.5)(0.02) = 0.03 \text{ m}^2$$

Part 3 — Ratio

$$\frac{A_e}{A_p} = \frac{1.17}{0.03} = \boxed{39.2}$$

Sanity: the antenna's "catching area" is $39\times$ its physical wire cross-section. A thin wire interacts with the field over a region $\sim \lambda$ around itself, so it captures much more energy than its tiny wire would suggest.

Problem 5 — Friis Transmission, TV Broadcast

5 pts

Setup: Half-wave dipole TX, $P_t = 1$ kW at $f = 50$ MHz. RX antenna with $G_r = 3$ dB, distance $R = 30$ km. Find P_r .

FRIIS TRANSMISSION FORMULA

For each antenna's G : if given by NAME ("half-wave dipole" etc.) $\rightarrow$ COMMON DIPOLE PARAMS ($G = D$, lossless). Else (dB/linear value given) $\rightarrow$ use as given (convert dB $\rightarrow$ linear).

Pwr density at RX, then RX pwr G_t, G_r linear gains;
 $\lambda = c/f$

$$S_r = \frac{G_t P_t}{4\pi R^2} \text{ (W/m}^2\text{)}; \quad P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t \text{ (W)}$$

Equivalent: $P_r = S_r A_{e,r}$.

Solve for R max range

$$R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}} \text{ (m)}$$

Equivalent form using A_e recv. area form

$$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, so $P_r = G_t G_r P_t / L_{fs}$.
 Off-axis (patterns not aligned): multiply by $F_t(\theta_t, \phi_t) F_r(\theta_r, \phi_r)$, both = 1 when aligned.

Recipe 1. $\lambda = c/f$. 2. Convert any dB gain $\rightarrow$ linear:
 $G = 10^{G_{dB}/10}$.

3. Get G_t, G_r by antenna type: dipole $\rightarrow$ COMMON DIPOLE PARAMS; aperture $\rightarrow$ APERTURE box; arbitrary $F \rightarrow$ BEAM PARAMS. 4. Plug into Friis.

Use linear G , not dB. Convert FIRST.

dB $\leftrightarrow$ LINEAR

Works for any power-like quantity (G , D , P_r/P_t , SNR, ...):

$$X_{dB} = 10 \log_{10} X_{lin} \quad X_{lin} = 10^{X_{dB}/10}$$

Memorize: 3 dB $\Leftrightarrow \times 2$, 10 dB $\Leftrightarrow \times 10$. Add dB = multiply linear.

For E-field / amplitude use 20 log not 10 log (power $\propto E^2$).

COMMON DIPOLE PARAMETERS

Lossless ($\xi = 1$): $G = D$.

Type	l	D	D_{dB}	$R_{rad} (\Omega)$
Isotropic	N/A	1	0	—
Hertzian	$< \lambda/50$	1.5	1.76	$80\pi^2 (l/\lambda)^2$
Half-wave	$\lambda/2$	1.64	2.15	73.2
Monopole	$\lambda/4$ (gnd)	1.64	2.15	36.6
Full-wave	λ	2.41	3.82	199

Half-wave: $P_{rad} = 36.6 I_0^2$ W. Monopole ($\lambda/4$ above ground): same patt. as half-wave but P_{rad}, R_{rad} halved.

Goal: find P_r using the Friis transmission formula (from FRIIS TRANSMISSION FORMULA box):

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t$$

Need: G_t, G_r (both linear), λ, R, P_t .

Solve each unknown:

λ : $\lambda = c/f$.

$$\lambda = \frac{3 \times 10^8}{5 \times 10^7} = 6 \text{ m}$$

G_t : TX is half-wave dipole, COMMON DIPOLE PARAMS gives $D = 1.64$, and for lossless antenna $G = D$.

$$G_t = 1.64$$

G_r : given as 3 dB. Convert to linear via dB $\rightarrow$ LINEAR table: 3 dB = $\times 2$.

$$G_r = 10^{3/10} = 2$$

R, P_t : given.

$$R = 30 \text{ km} = 3 \times 10^4 \text{ m}, \quad P_t = 10^3 \text{ W}$$

Plug back in: compute the path-loss factor first.

Step 1 — compute $\lambda/(4\pi R)$:

$$4\pi R = 4\pi(3 \times 10^4) = 4(3.14159)(3 \times 10^4) = 37.70 \times 10^4 = 3.770 \times 10^5$$

$$\frac{\lambda}{4\pi R} = \frac{6}{3.770 \times 10^5} = 1.591 \times 10^{-5}$$

Step 2 — square it:

$$\left(\frac{\lambda}{4\pi R}\right)^2 = (1.591 \times 10^{-5})^2 = 2.531 \times 10^{-10}$$

Step 3 — multiply by $G_t G_r$:

$$G_t G_r = 1.64 \cdot 2 = 3.28$$

$$G_t G_r \cdot \left(\frac{\lambda}{4\pi R}\right)^2 = 3.28 \cdot 2.531 \times 10^{-10} = 8.30 \times 10^{-10}$$

Step 4 — multiply by $P_t = 10^3 \text{ W}$:

$$P_r = 8.30 \times 10^{-10} \cdot 10^3 = 8.30 \times 10^{-7} \text{ W}$$

Convert to readable unit: $10^{-6} = \text{micro } (\mu)$, so $8.30 \times 10^{-7} = 0.830 \times 10^{-6} \text{ W}$:

$$P_r = 8.30 \times 10^{-7} \text{ W} = 0.830 \mu\text{W} = 830 \text{ nW}$$

Sanity: 1 kW in, 0.83 μW out at 30 km. A 90 dB drop, dominated by $1/R^2$ path loss. Plenty for a TV receiver (sensitivity $\sim 10^{-13} \text{ W}$).

Problem 6 — Communication Link with SNR

15 pts (new)

Setup: $G_t = 20$ dB, $G_r = 23$ dB, $R = 20$ km, $P_t = 10$ W, $f = 6$ GHz. (c) $T_{\text{sys}} = 1000$ K, $B = 20$ MHz.

FRIIS TRANSMISSION FORMULA

For each antenna's G : if given by NAME ("half-wave dipole" etc.) $\rightarrow$ COMMON DIPOLE PARAMS ($G = D$, lossless). Else (dB/linear value given) $\rightarrow$ use as given (convert dB $\rightarrow$ linear).

Pwr density at RX, then RX pwr G_t, G_r linear gains;
 $\lambda = c/f$

$$S_r = \frac{G_t P_t}{4\pi R^2} \text{ (W/m}^2\text{)}; \quad P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t \text{ (W)}$$

Equivalent: $P_r = S_r A_{e,r}$.

Solve for R max range

$$R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}} \text{ (m)}$$

Equivalent form using A_e recv. area form

$$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, so $P_r = G_t G_r P_t / L_{fs}$.
 Off-axis (patterns not aligned): multiply by $F_t(\theta_t, \phi_t) F_r(\theta_r, \phi_r)$, both = 1 when aligned.

Recipe 1. $\lambda = c/f$. 2. Convert any dB gain $\rightarrow$ linear:
 $G = 10^{G_{\text{dB}}/10}$.

3. Get G_t, G_r by antenna type: dipole $\rightarrow$ COMMON DIPOLE PARAMS; aperture $\rightarrow$ APERTURE box; arbitrary $F \rightarrow$ BEAM FOCUS PARAMS. 4. Plug into Friis.

Use linear G , not dB. Convert FIRST.

dB $\leftrightarrow$ LINEAR

Works for any power-like quantity (G , D , P_r/P_t , SNR, ...):

$$X_{\text{dB}} = 10 \log_{10} X_{\text{lin}} \quad X_{\text{lin}} = 10^{X_{\text{dB}}/10}$$

Memorize: 3 dB $\leftrightarrow \times 2$, 10 dB $\leftrightarrow \times 10$. Add dB = multiply linear.

For E-field / amplitude use 20 log not 10 log (power $\propto E^2$).

NOISE & SNR

Noise power thermal noise into matched receiver

$$P_N = k_B T_{\text{sys}} B \text{ (W)}$$

$k_B = 1.38 \times 10^{-23}$ J/K; T_{sys} system noise temp (K); B receiver bandwidth (Hz).

Signal-to-noise ratio

$$\text{SNR} = \frac{P_{\text{rec}}}{P_N} \text{ (unitless)} \quad \text{SNR}_{\text{dB}} = 10 \log_{10} \frac{P_{\text{rec}}}{P_N}$$

Shared setup: convert dB to linear and compute λ .

Wavelength:

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8 \text{ m/s}}{6 \times 10^9 \text{ Hz}} = 0.05 \text{ m} = 5 \text{ cm}$$

Convert $G_t = 20$ dB to linear (from dB $\leftrightarrow$ LINEAR box: $X_{\text{lin}} = 10^{X_{\text{dB}}/10}$):

$$G_t = 10^{20/10} = 10^2 = 100 \text{ (unitless)}$$

Convert $G_r = 23$ dB to linear:

$$G_r = 10^{23/10} = 10^{2.3} = 199.5 \text{ (unitless)}$$

(a) Power density at RX antenna S_r (5 pts)

Goal: find S_r using power-spreading-over-sphere (with TX gain factored in):

$$S_r = \frac{G_t P_t}{4\pi R^2} \text{ (W/m}^2\text{)}$$

Need: G_t, P_t, R .

Plug in step by step:

Numerator: $G_t P_t = 100 \cdot 10 = 1000 \text{ W}$

Square R: $R^2 = (2 \times 10^4)^2 = 4 \times 10^8 \text{ m}^2$

Denominator: $4\pi R^2 = 4\pi(4 \times 10^8) = 4(3.14159)(4 \times 10^8) = 50.27 \times 10^8 = 5.027 \times 10^9 \text{ m}^2$

Divide:

$$S_r = \frac{1000}{5.027 \times 10^9} = 1.989 \times 10^{-7} \text{ W/m}^2$$

Convert to readable unit: $10^{-6} = \text{micro } (\mu)$, so $1.989 \times 10^{-7} = 0.1989 \times 10^{-6}$:

$$S_r = 1.989 \times 10^{-7} \text{ W/m}^2 \approx 0.199 \mu\text{W/m}^2$$

(b) Received power P_r (5 pts)

Goal: find P_r using Friis (from FRIIS TRANSMISSION box):

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t$$

Need: G_t , G_r , λ , R , P_t (all already converted).

Plug in:

$$\frac{\lambda}{4\pi R} = \frac{0.05}{4\pi(2 \times 10^4)} = 1.989 \times 10^{-7}$$

$$\left(\frac{\lambda}{4\pi R} \right)^2 = 3.957 \times 10^{-14}$$

$$P_r = (100)(199.5)(3.957 \times 10^{-14})(10) = 7.894 \times 10^{-9}$$

$$P_r = 7.89 \text{ nW}$$

Cross-check via effective area: $A_{e,r} = G_r \lambda^2 / (4\pi) = 199.5(0.0025) / (4\pi) = 0.0397 \text{ m}^2$, then $P_r = S_r A_{e,r} = (1.989 \times 10^{-7})(0.0397) = 7.89 \times 10^{-9} \text{ W}$ ✓.

(c) Signal-to-noise ratio in dB (5 pts)

Goal: find SNR in dB. From NOISE & SNR box, two steps:

$$P_N = k_B T_{\text{sys}} B \quad \text{then} \quad \text{SNR}_{\text{dB}} = 10 \log_{10} \frac{P_r}{P_N}$$

Need: P_N first ($\rightarrow$ need k_B , T_{sys} , B), then P_r (from part b).

Solve each unknown:

k_B : Boltzmann constant.

$$k_B = 1.38 \times 10^{-23} \text{ J/K}$$

T_{sys} , B : given.

$$T_{\text{sys}} = 1000 \text{ K}, \quad B = 20 \text{ MHz} = 2 \times 10^7 \text{ Hz}$$

Compute P_N :

$$P_N = (1.38 \times 10^{-23})(1000)(2 \times 10^7) = 2.76 \times 10^{-13} \text{ W}$$

Compute SNR:

$$\text{SNR}_{\text{lin}} = \frac{P_r}{P_N} = \frac{7.89 \times 10^{-9}}{2.76 \times 10^{-13}} = 2.86 \times 10^4$$

$$\text{SNR}_{\text{dB}} = 10 \log_{10}(2.86 \times 10^4) = 44.6 \text{ dB}$$

Sanity: $>40 \text{ dB}$ SNR is a very clean link. Plenty for digital comm.

Problem 7 — Circular Aperture Antenna Scaling

15 pts (new)

Setup: Circular aperture, circular beam $\beta = 3^\circ$ at $f = 20$ GHz. (a) D in dB. (b) New D and β if A_p doubled. (c) New D and β if frequency doubled (same A_p).

APERTURE ANTENNAS (horn, dish)

A_p physical aperture area (m^2); A_e effective area (m^2 , $= \lambda^2 D / 4\pi$); β_{xz}, β_{yz} HPBWs in two planes (rad); ℓ , d aperture side / diameter (m).

Far-field req.: $R \geq 2d^2 / \lambda$ (d = largest aperture dim.).

Pattern (rect., uniform illum.) peak at $\theta = 0$ (broadside)

$$F(\theta) = \text{sinc}^2\left(\frac{\pi \ell_x \sin \theta}{\lambda}\right); \quad S_0 = \frac{E_0^2 A_p^2}{2\eta_0 \lambda^2 R^2}$$

Directivity (any shape, uniform illum.)

$$D \approx \frac{4\pi A_p}{\lambda^2} \text{ (unitless)} \Rightarrow A_e \approx A_p$$

Directivity from beamwidths any aperture

$$D \approx \frac{4\pi}{\beta_{xz}\beta_{yz}} \text{ (use rad); circular: } D \approx 4\pi/\beta^2$$

HPBW and D by aperture shape uniform illum.; β in rad

Shape	A_p	HPBW (rad)	D (unitless)
Rect. $\ell_x \times \ell_y$	$\ell_x \ell_y$	$\beta_x \approx 0.88\lambda/\ell_x$ $\beta_y \approx 0.88\lambda/\ell_y$	$D \approx 4\pi/(\beta_x \beta_y)$ $= 4\pi \ell_x \ell_y / \lambda^2$
Square ℓ	ℓ^2	$\beta \approx 0.88\lambda/\ell$	$D \approx 4\pi/\beta^2 = 4\pi \ell^2 / \lambda^2$
Circular (diam. d)	$\pi d^2/4$	$\beta \approx 1.02\lambda/d$	$D \approx 4\pi/\beta^2 = \pi^2 d^2 / \lambda^2$

Scaling (any ratio) $D \propto A_p / \lambda^2$; $\beta \propto \lambda / \sqrt{A_p}$

$$D_{\text{new}} = D_{\text{old}} \cdot \frac{A_{p,\text{new}}}{A_{p,\text{old}}} \cdot \left(\frac{f_{\text{new}}}{f_{\text{old}}}\right)^2$$

$$\beta_{\text{new}} = \beta_{\text{old}} \cdot \sqrt{\frac{A_{p,\text{old}}}{A_{p,\text{new}}}} \cdot \frac{f_{\text{old}}}{f_{\text{new}}}$$

If a parameter is unchanged, its ratio = 1. Examples: double area $\rightarrow D \times 2$, $\beta \times 1/\sqrt{2}$; double freq $\rightarrow D \times 4$, $\beta \times 1/2$.

dB $\leftrightarrow$ LINEAR

Works for any power-like quantity (G , D , P_r/P_t , SNR , ...):

$$X_{\text{dB}} = 10 \log_{10} X_{\text{lin}} \quad X_{\text{lin}} = 10^{X_{\text{dB}}/10}$$

Memorize: 3 dB $\Leftrightarrow \times 2$, 10 dB $\Leftrightarrow \times 10$. Add dB = multiply linear.

For E-field / amplitude use 20 log not 10 log (power $\propto E^2$).

(a) Directivity in dB (5 pts)

Goal: find D using the circular-beam formula (from APERTURE ANTENNAS box):

$$D = \frac{4\pi}{\beta^2} \quad (\beta \text{ in radians})$$

Need: β in radians (given as degrees).

Convert β to radians (deg $\rightarrow$ rad, multiply by $\pi/180$, from COMMON ANGLES box):

$$\begin{aligned} \beta &= 3^\circ \cdot \frac{\pi}{180} = \frac{3\pi}{180} = \frac{\pi}{60} \\ &= \frac{3.14159}{60} = 0.05236 \text{ rad} \end{aligned}$$

Plug in:

$$D = \frac{4\pi}{(0.05236)^2} = \frac{12.566}{0.002742} = 4584 \text{ (unitless)}$$

Convert to dB using the standard linear $\rightarrow$ dB formula (from dB $\leftrightarrow$ LINEAR box):

$$D_{\text{dB}} = 10 \log_{10}(D_{\text{lin}}) = 10 \log_{10}(4584) = 10 \cdot 3.661$$

$$D_{\text{dB}} = 36.6 \text{ dB}$$

(b) Area doubled $A_p \rightarrow 2A_p$ (5 pts)

Goal: find new D and β using the scaling rules (from APERTURE ANTENNAS box, “Scaling (any ratio)” block):

$$D_{\text{new}} = D_{\text{old}} \cdot \frac{A_{p,\text{new}}}{A_{p,\text{old}}} \cdot \left(\frac{f_{\text{new}}}{f_{\text{old}}} \right)^2, \quad \beta_{\text{new}} = \beta_{\text{old}} \cdot \sqrt{\frac{A_{p,\text{old}}}{A_{p,\text{new}}}} \cdot \frac{f_{\text{old}}}{f_{\text{new}}}$$

Here f unchanged, $A_p \rightarrow 2A_p$, so $D \propto A_p$ and $\beta \propto 1/\sqrt{A_p}$:

$$D_{\text{new}} = 2D_{\text{old}}, \quad \beta_{\text{new}} = \beta_{\text{old}}/\sqrt{2}$$

Need: D_{old} and β_{old} (from part a).

Plug in:

$$D_{\text{new}} = 2(4584) = 9168 \Rightarrow \boxed{D_{\text{new}} = 39.6 \text{ dB}}$$

$$\beta_{\text{new}} = 3^\circ/\sqrt{2} = 3^\circ/1.414 = \boxed{2.12^\circ}$$

(c) **Frequency doubled** $f \rightarrow 2f$ ($\lambda \rightarrow \lambda/2$), same A_p (5 pts)

Goal: find new D and β using scaling: from $D = 4\pi A_p/\lambda^2$ we have $D \propto 1/\lambda^2$. Halving λ quadruples D . And $\beta \propto \lambda$, so halving λ halves β . **Need:** D_{old} and β_{old} (from part a).

Plug in:

$$D_{\text{new}} = 4D_{\text{old}} = 4(4584) = 18336 \Rightarrow \boxed{D_{\text{new}} = 42.6 \text{ dB}}$$

$$\beta_{\text{new}} = \beta_{\text{old}}/2 = 3^\circ/2 = \boxed{1.5^\circ}$$

Sanity: bigger area or shorter wavelength both sharpen the beam. Doubling area: +3 dB ($\times 2$). Doubling frequency: +6 dB ($\times 4$) because both $D \propto A_p/\lambda^2$ and $1/\beta^2 \propto D$.

Problem 8 — 5-Element Linear Array, $d = 3\lambda/4$

(new)

Setup: Broadside linear array, $N = 5$ elements, equal-phase, uniform amplitude, spacing $d = 3\lambda/4$. Find (a) normalized array factor, (b) HPBW.

LINEAR ANTENNA ARRAY

Array factor (uniform, broadside) $k = 2\pi/\lambda$, $\gamma = kd \cos \theta$

$$F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}; \quad F_a^{\max} = N^2 \text{ at } \theta = \pi/2$$

Normalize $F_a^{\text{norm}} = F_a/N^2$

HPBW (broadside, uniform) *use Nd , not $(N-1)d$*

$$\beta \approx 0.88 \frac{\lambda}{Nd} \text{ (rad)} = 50.8^\circ \cdot \frac{\lambda}{Nd}$$

Electronic steering (scan to θ_0) *replace γ with $\gamma' = kd(\cos \theta - \cos \theta_0)$*

$$\delta = \frac{2\pi d}{\lambda} \cos \theta_0 \left(\frac{\text{rad}}{\text{elt}} \right); \quad \theta_0 = \frac{\pi}{2} : \text{broadside}, 0 : \text{end-fire}$$

Frequency scan *feed-line incr. $\Delta \ell = n_0 \lambda_0$*

$$\cos \theta_0 \approx \frac{n_0 \lambda_0}{d} \frac{\Delta f}{f_0}$$

Grating lobes appear if $d > \lambda$. Avoid.

(a) Normalized array factor $F_a^{\text{norm}}(\theta)$ (5 pts)

Goal: find $F_a^{\text{norm}}(\theta)$ using the uniform-array closed form (from LINEAR ANTENNA ARRAY box):

$$F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}, \quad \gamma = kd \cos \theta$$

Need: N , k , d , then form γ , then normalize by peak N^2 .

Solve each unknown:

N : given.

$$N = 5$$

k : wavenumber.

$$k = \frac{2\pi}{\lambda}$$

d : given.

$$d = \frac{3\lambda}{4}$$

γ : combine.

$$\gamma = kd \cos \theta = \frac{2\pi}{\lambda} \cdot \frac{3\lambda}{4} \cos \theta = \frac{3\pi}{2} \cos \theta$$

So $N\gamma/2 = (5/2)(3\pi/2) \cos \theta = (15\pi/4) \cos \theta$ and $\gamma/2 = (3\pi/4) \cos \theta$.

Plug into F_a :

$$F_a(\theta) = \frac{\sin^2((15\pi/4) \cos \theta)}{\sin^2((3\pi/4) \cos \theta)}$$

Peak occurs at $\gamma = 0$ ($\cos \theta = 0$, broadside $\theta = \pi/2$): $F_a^{\max} = N^2 = 25$.

Normalize:

$$F_a^{\text{norm}}(\theta) = \frac{1}{25} \frac{\sin^2((15\pi/4) \cos \theta)}{\sin^2((3\pi/4) \cos \theta)}$$

(b) Half-power beamwidth β (3 pts)

Goal: find β using the uniform-broadside-array HPBW formula:

$$\beta \approx 0.886 \frac{\lambda}{Nd} \quad (\text{rad})$$

Need: N , d , λ .

Plug in:

$$Nd = 5 \cdot \frac{3\lambda}{4} = \frac{15\lambda}{4}$$

$$\beta = 0.886 \cdot \frac{\lambda}{15\lambda/4} = 0.886 \cdot \frac{4}{15} = 0.236 \text{ rad}$$

$$\boxed{\beta = 0.236 \text{ rad} = 13.5^\circ}$$

Cross-check by solving $F_a^{\text{norm}} = 0.5$ numerically: get $\gamma/2 \approx 0.28$ rad, so $\cos \theta = 0.28/(3\pi/4) = 0.119$, $\theta = 83.2^\circ$, offset from broadside (90°) by 6.8° , full HPBW $\approx 13.6^\circ$ ✓.

Sanity: 5 elements give a moderately narrow beam ($\sim 14^\circ$) vs. a single dipole's $\sim 78^\circ$. Adding more elements or wider spacing narrows further (with grating-lobe risk if $d > \lambda$).